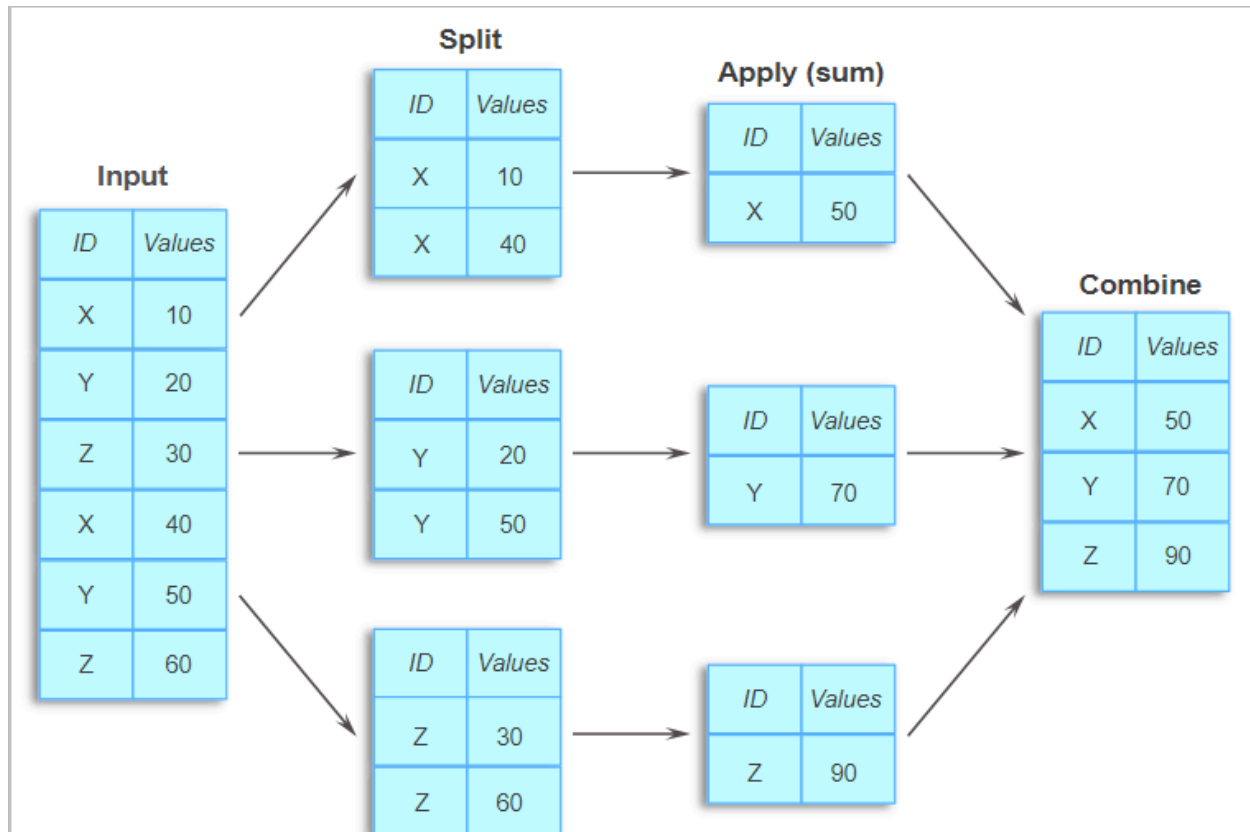




1. Write a Pandas program to split the following data frame into groups based on school code. Also check the type of GroupBy object.

Test Data:

	school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35	street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32	street2
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31	street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32	street4

2. Write a Pandas program to split the following dataframe by school code and get mean, min, and max value of age for each school.

Test Data:

	school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35	street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32	street2

S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31	street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32	street4

3. Write a Pandas program to split the following given dataframe into groups based on school code and class.

Test Data:

	school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35	street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32	street2
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31	street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32	street4

4. Write a Pandas program to split the following given dataframe into groups based on school code and cast grouping as a list.

Test Data:

	school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35	street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32	street2
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31	street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32	street4

5. Write a Pandas program to split the following given dataframe into groups based on single column and multiple columns. Find the size of the grouped data.

Test Data:

	school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35	street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32	street2
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31	street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32	street4

6. Write a Pandas program to split the following given dataframe into groups based on school code and call a specific group with the name of the group.

Test Data:

	school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35	street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32	street2
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31	street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32	street4

7. Write a Pandas program to split a dataset, group by one column and get mean, min, and max values by group.

Using the following dataset find the mean, min, and max values of purchase amount (purch_amt) group by customer id (customer_id).

Test Data:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	2012-10-05	3005	5002
1	70009	270.65	2012-09-10	3001	5005
2	70002	65.26	2012-10-05	3002	5001
3	70004	110.50	2012-08-17	3009	5003
4	70007	948.50	2012-09-10	3005	5002
5	70005	2400.60	2012-07-27	3007	5001
6	70008	5760.00	2012-09-10	3002	5001
7	70010	1983.43	2012-10-10	3004	5006
8	70003	2480.40	2012-10-10	3009	5003
9	70012	250.45	2012-06-27	3008	5002
10	70011	75.29	2012-08-17	3003	5007
11	70013	3045.60	2012-04-25	3002	5001

8. Write a Pandas program to split a dataset to group by two columns and count by each row.

Test Data:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	2012-10-05	3005	5002
1	70009	270.65	2012-09-10	3001	5005
2	70002	65.26	2012-10-05	3002	5001
3	70004	110.50	2012-08-17	3009	5003

4	70007	948.50	2012-09-10	3005	5002
5	70005	2400.60	2012-07-27	3007	5001
6	70008	5760.00	2012-09-10	3002	5001
7	70010	1983.43	2012-10-10	3004	5006
8	70003	2480.40	2012-10-10	3009	5003
9	70012	250.45	2012-06-27	3008	5002
10	70011	75.29	2012-08-17	3003	5007
11	70013	3045.60	2012-04-25	3002	5001

9. Write a Pandas program to split a dataset to group by two columns and then sort the aggregated results within the groups.

In the following dataset group on 'customer_id', 'salesman_id' and then sort sum of purch_amt within the groups.

Test Data:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	2012-10-05	3005	5002
1	70009	270.65	2012-09-10	3001	5005
2	70002	65.26	2012-10-05	3002	5001
3	70004	110.50	2012-08-17	3009	5003
4	70007	948.50	2012-09-10	3005	5002
5	70005	2400.60	2012-07-27	3007	5001
6	70008	5760.00	2012-09-10	3002	5001
7	70010	1983.43	2012-10-10	3004	5006
8	70003	2480.40	2012-10-10	3009	5003
9	70012	250.45	2012-06-27	3008	5002
10	70011	75.29	2012-08-17	3003	5007
11	70013	3045.60	2012-04-25	3002	5001

10. Write a Pandas program to split the following dataframe into groups based on customer id and create a list of order date for each group.

Test Data:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	2012-10-05	3005	5002
1	70009	270.65	2012-09-10	3001	5005
2	70002	65.26	2012-10-05	3002	5001
3	70004	110.50	2012-08-17	3009	5003
4	70007	948.50	2012-09-10	3005	5002
5	70005	2400.60	2012-07-27	3007	5001
6	70008	5760.00	2012-09-10	3002	5001
7	70010	1983.43	2012-10-10	3004	5006

8	70003	2480.40	2012-10-10	3009	5003
9	70012	250.45	2012-06-27	3008	5002
10	70011	75.29	2012-08-17	3003	5007
11	70013	3045.60	2012-04-25	3002	5001

11. Write a Pandas program to split the following dataframe into groups and calculate monthly purchase amount.

Test Data:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	05-10-2012	3001	5002
1	70009	270.65	09-10-2012	3001	5005
2	70002	65.26	05-10-2012	3005	5001
3	70004	110.50	08-17-2012	3001	5003
4	70007	948.50	10-09-2012	3005	5002
5	70005	2400.60	07-27-2012	3001	5001
6	70008	5760.00	10-09-2012	3005	5001
7	70010	1983.43	10-10-2012	3001	5006
8	70003	2480.40	10-10-2012	3005	5003
9	70012	250.45	06-17-2012	3001	5002
10	70011	75.29	07-08-2012	3005	5007
11	70013	3045.60	04-25-2012	3005	5001

12. Write a Pandas program to split the following dataframe into groups, group by month and year based on order date and find the total purchase amount year wise, month wise.

Test Data:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	05-10-2012	3001	5002
1	70009	270.65	09-10-2012	3001	5005
2	70002	65.26	05-10-2012	3005	5001
3	70004	110.50	08-17-2012	3001	5003
4	70007	948.50	10-09-2012	3005	5002
5	70005	2400.60	07-27-2012	3001	5001
6	70008	5760.00	10-09-2012	3005	5001
7	70010	1983.43	10-10-2012	3001	5006
8	70003	2480.40	10-10-2012	3005	5003
9	70012	250.45	06-17-2012	3001	5002
10	70011	75.29	07-08-2012	3005	5007
11	70013	3045.60	04-25-2012	3005	5001

13. Write a Pandas program to split the following dataframe into groups based on first column and set other column values into a list of values.

Test Data:

```
   X  Y  Z
0 10 10 22
1 10 15 20
2 10 11 18
3 20 20 20
4 30 21 13
5 30 12 10
6 10 14  0
```

14. Write a Pandas program to split the following dataframe into groups based on all columns and calculate Groupby value counts on the dataframe.

Test Data:

```
   id type  book
0  1  10  Math
1  2  15 English
2  1  11 Physics
3  1  20  Math
4  2  21 English
5  1  12 Physics
6  2  14 English
```

Output:

```
book  English  Math  Physics id type
1 10         0    1     0
   11         0    0     1
   12         0    0     1
   20         0    1     0
2 14         1    0     0
   15         1    0     0
   21         1    0     0
```

15. Write a Pandas program to split the following dataframe into groups and count unique values of 'value' column.

Test Data:

```
   id value
0  1     a
```

```

1 1 a
2 2 b
3 3 None
4 3 a
5 4 a
6 4 None
7 4 b

```

Output:

```

value
a 3
b 2

```

16. Write a Pandas program to split a given dataframe into groups and list all the keys from the GroupBy object.

Test Data:

	school_code	class	name	date_Of_Birth	age	height	weight
S1	s001	V	Alberto Franco	15/05/2002	12	173	35
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31
S6	s004	VI	David Parkes	15/09/1997	12	159	32

17. Write a Pandas program to split a given dataframe into groups and create a new column with count from GroupBy.

Test Data:

	book_name	book_type	book_id
0	Book1	Math	1
1	Book2	Physics	2
2	Book3	Computer	3
3	Book4	Science	4
4	Book1	Math	1
5	Book2	Physics	2
6	Book3	Computer	3
7	Book5	English	5

18. Write a Pandas program to split a given dataframe into groups with bin counts.

Test Data:

	ord_no	purch_amt	customer_id	sales_id
0	70001	150.50	3005	5002
1	70009	270.65	3001	5003
2	70002	65.26	3002	5004
3	70004	110.50	3009	5003
4	70007	948.50	3005	5002
5	70005	2400.60	3007	5001
6	70008	5760.00	3002	5005
7	70010	1983.43	3004	5007
8	70003	2480.40	3009	5008
9	70012	250.45	3008	5004
10	70011	75.29	3003	5005
11	70013	3045.60	3002	5001

19. Write a Pandas program to split a given dataframe into groups with multiple aggregations.

Split the following given dataframe by school code, class and get mean, min, and max value of height and age for each value of the school.

Test Data:

	school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35	street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32	street2
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31	street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32	street4

20. Write a Pandas program to split a given dataframe into groups and display target column as a list of unique values.

Test Data:

	id	type	book
0	A	1	Math
1	A	1	Math
2	A	1	English
3	A	1	Physics
4	A	2	Math
5	A	2	English
6	B	1	Physics
7	B	1	English
8	B	1	Physics

9 B 2 English
10 B 2 English

Output:

List all unique values in a group:

	id	type	book
0	A	1	Math,English,Physics
1	A	2	Math,English
2	B	1	Physics,English
3	B	2	English

21. Write a Pandas program to split the following dataframe into groups and calculate quarterly purchase amount.

Test Data:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	05-10-2012	3001	5002
1	70009	270.65	09-10-2012	3001	5005
2	70002	65.26	05-10-2012	3005	5001
3	70004	110.50	08-17-2012	3001	5003
4	70007	948.50	10-09-2012	3005	5002
5	70005	2400.60	07-27-2012	3001	5001
6	70008	5760.00	10-09-2012	3005	5001
7	70010	1983.43	10-10-2012	3001	5006
8	70003	2480.40	10-10-2012	3005	5003
9	70012	250.45	06-17-2012	3001	5002
10	70011	75.29	07-08-2012	3005	5007
11	70013	3045.60	04-25-2012	3005	5001

22. Write a Pandas program to split the following dataframe into groups by school code and get mean, min, and max value of age with customized column name for each school.

Test Data:

	school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35	street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32	street2
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31	street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32	street4

23. Write a Pandas program to split the following datasets into groups on customer id and calculate the number of customers starting with 'C', the list of all products and the difference of maximum purchase amount and minimum purchase amount.

Test Data:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	05-10-2012	C3001	5002
1	70009	270.65	09-10-2012	C3001	5005
2	70002	65.26	05-10-2012	D3005	5001
3	70004	110.50	08-17-2012	D3001	5003
4	70007	948.50	10-09-2012	C3005	5002
5	70005	2400.60	07-27-2012	D3001	5001
6	70008	5760.00	10-09-2012	C3005	5001
7	70010	1983.43	10-10-2012	D3001	5006
8	70003	2480.40	10-10-2012	D3005	5003
9	70012	250.45	06-17-2012	C3001	5002
10	70011	75.29	07-08-2012	D3005	5007
11	70013	3045.60	04-25-2012	D3005	5001

24. Write a Pandas program to split the following datasets into groups on customer_id to summarize purch_amt and calculate percentage of purch_amt in each group.

Test Data:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	05-10-2012	3001	5002
1	70009	270.65	09-10-2012	3001	5005
2	70002	65.26	05-10-2012	3005	5001
3	70004	110.50	08-17-2012	3001	5003
4	70007	948.50	10-09-2012	3005	5002
5	70005	2400.60	07-27-2012	3001	5001
6	70008	5760.00	10-09-2012	3005	5001
7	70010	1983.43	10-10-2012	3001	5006
8	70003	2480.40	10-10-2012	3005	5003
9	70012	250.45	06-17-2012	3001	5002
10	70011	75.29	07-08-2012	3005	5007
11	70013	3045.60	04-25-2012	3005	5001

25. Write a Pandas program to split a dataset, group by one column and get mean, min, and max values by group, also change the column name of the aggregated metric. Using the following dataset find the mean, min, and max values of purchase amount (purch_amt) group by customer id (customer_id).

Test Data:

	school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35	street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32	street2
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31	street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32	street4

26. Write a Pandas program to split a given dataset, group by two columns and convert other columns of the dataframe into a dictionary with column header as key.

Test Data:

	school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35	street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32	street2
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31	street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32	street4

27. Write a Pandas program to split a given dataset, group by one column and apply an aggregate function to few columns and another aggregate function to the rest of the columns of the dataframe.

Test Data:

	salesman_id	sale_jan	sale_feb	sale_mar	sale_apr	sale_may	sale_jun \
0	5002	150.50	250.50	150.50	150.50	130.50	150.50
1	5005	270.65	170.65	270.65	270.65	270.65	270.65
2	5001	65.26	15.26	65.26	95.26	65.26	45.26
3	5003	110.50	110.50	110.50	210.50	310.50	110.50
4	5002	948.50	598.50	948.50	948.50	948.50	948.50
5	5001	2400.60	1400.60	2400.60	2400.60	2400.60	3400.60
6	5001	1760.00	2760.00	5760.00	760.00	760.00	5760.00
7	5006	2983.43	1983.43	1983.43	1983.43	1983.43	983.43
8	5003	480.40	2480.40	2480.40	2480.40	2480.40	2480.40
9	5002	1250.45	250.45	250.45	250.45	250.45	250.45
10	5007	75.29	75.29	75.29	75.29	75.29	75.29
11	5001	1045.60	3045.60	3045.60	3045.60	3045.60	3045.60

	sale_jul	sale_aug	sale_sep	sale_oct	sale_nov	sale_dec
0	950.50	150.50	150.50	150.50	150.50	150.50

1	270.65	70.65	270.65	270.65	270.65	70.65
2	65.26	65.26	65.26	65.26	95.26	65.26
3	210.50	110.50	110.50	110.50	110.50	110.50
4	948.50	948.50	948.50	948.50	948.50	948.50
5	2400.60	400.60	200.60	2400.60	2400.60	2400.60
6	5760.00	5760.00	5760.00	5760.00	5760.00	5760.00
7	983.43	1983.43	1983.43	1983.43	1983.43	1983.43
8	2480.40	2480.40	2480.40	2480.40	2480.40	2480.40
9	250.45	250.45	250.45	250.45	250.45	250.45
10	75.29	75.29	75.29	75.29	75.29	75.29
11	3045.60	3045.60	3045.60	3045.60	3045.60	3045.60

28. Write a Pandas program to split a given dataset, group by one column and remove those groups if all the values of a specific columns are not available.

Test Data:

	school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35	street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32	street2
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31	street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32	street4

29. Write a Pandas program to split a given dataset using group by on specified column into two labels and ranges.

Split the group on 'salesman_id',

Ranges:

1) (5001...5006)

2) (5007..5012)

Test Data:

	salesman_id	sale_jan
0	5001	150.50
1	5002	270.65
2	5003	65.26
3	5004	110.50
4	5005	948.50
5	5006	2400.60

6	5007	1760.00
7	5008	2983.43
8	5009	480.40
9	5010	1250.45
10	5011	75.29
11	5012	1045.60

30. Write a Pandas program to split the following dataset using group by on first column and aggregate over multiple lists on second column.

Test Data:

	student_id	marks
0	S001	[88, 89, 90]
1	S001	[78, 81, 60]
2	S002	[84, 83, 91]
3	S002	[84, 88, 91]
4	S003	[90, 89, 92]
5	S003	[88, 59, 90]

Output:

student_id	marks
S001	[83.0, 85.0, 75.0]
S002	[84.0, 85.5, 91.0]
S003	[89.0, 74.0, 91.0]

31. Write a Pandas program to split the following dataset using group by on 'salesman_id' and find the first order date for each group.

Test Data:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	2012-10-05	3005	5002
1	70009	270.65	2012-09-10	3001	5005
2	70002	65.26	2012-10-05	3002	5001
3	70004	110.50	2012-08-17	3009	5003
4	70007	948.50	2012-09-10	3005	5002
5	70005	2400.60	2012-07-27	3007	5001
6	70008	5760.00	2012-09-10	3002	5001
7	70010	1983.43	2012-10-10	3004	5004
8	70003	2480.40	2012-10-10	3009	5003
9	70012	250.45	2012-06-27	3008	5002
10	70011	75.29	2012-08-17	3003	5004
11	70013	3045.60	2012-04-25	3002	5001

32. Write a Pandas program to split a given dataset using group by on multiple columns and drop last n rows of from each group.

Test Data:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	2012-10-05	3002	5002
1	70009	270.65	2012-09-10	3001	5003
2	70002	65.26	2012-10-05	3001	5001
3	70004	110.50	2012-08-17	3003	5003
4	70007	948.50	2012-09-10	3002	5002
5	70005	2400.60	2012-07-27	3002	5001
6	70008	5760.00	2012-09-10	3001	5001
7	70010	1983.43	2012-10-10	3004	5003
8	70003	2480.40	2012-10-10	3003	5003
9	70012	250.45	2012-06-27	3002	5002
10	70011	75.29	2012-08-17	3003	5003
11	70013	3045.60	2012-04-25	3001	5001